Burns

- Administer oxygen if the patient has suspected carbon monoxide poisoning as a result of clinically significant smoke inhalation. See the **'poisoning from gases'** guideline for more information.
- Cool the burn for at least 20 minutes:
  - This should be at the scene unless there is an immediately life-threatening problem in the primary survey.
  - Remove all clothing and decontaminate the patient if the burns are due to chemical exposure.
- Irrigate chemical burns to the eye for at least 30 minutes.
- Estimate burn depth and size after cooling.
- Cover burns with cling film after cooling.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia, or the burn area is greater than 20%:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Administer further 0.9% sodium chloride if transport time is greater than one hour, or if required for hypovolaemia.
- Administer nebulised bronchodilators (using the **'exacerbation of asthma'** guideline) if bronchospasm is prominent.
- Administer nebulised adrenaline (using the **'stridor'** guideline) if stridor is present.
- Consider RSI if there are immediate concerns for airway compromise, or if the burn extensively involves the face/head and transport time will be prolonged. Have a low threshold for seeking clinical advice.

Backup

- Backup from a CCP or should be requested if the patient has airway burns or is unable to obey commands.

Referral and transport

- Transport the patient to a regional burn centre if the burn size is greater than 10% of TBSA in an adult or greater than 5% of TBSA in a child, whenever it is feasible and safe to do so.
- The regional burn centres are Middlemore Hospital, Waikato Hospital, Hutt Hospital and Christchurch Hospital. Note that Hutt Hospital is not a major trauma hospital and if there are signs or symptoms of

major trauma in addition to the burn injury the patient must be transported to a major trauma hospital.

- If it is not feasible or safe to transport the patient to a regional burn centre, the patient should be transported to a major trauma hospital.
- Patients with burns involving the airway should usually be transported to a regional burns centre. However, if there are signs of airway compromise, the patient should be transported to the closest hospital with personnel able to perform RSI.
- Patients with burns involving the face, hands or genitals may require treatment in a regional burn centre. However, provided the burn size is less than 10% of TBSA in an adult or less than 5% of TBSA in a child, treatment is not usually time sensitive and the patient should usually be transported to the most appropriate hospital, and subsequently transferred if required.
- Patients with a burn secondary to an electrical injury should be transported to a major trauma hospital, even if the burn area is small.

Additional information

Cooling, irrigating and dressing burns

- Burns are preferably cooled using cool (not ice cold) running water:
  - There is benefit in cooling the burn up to three hours post injury.
  - Beware of hypothermia during cooling, particularly if the patient is a small child or the burn is large.
  - It is acceptable to shorten the duration of cooling for burns greater than 40% if it risks causing clinically significant hypothermia.
  - Cool the burn but keep the patient warm.
- Chemical burns of the eye are potentially vision-threatening and irrigation must continue for at least 30 minutes. Chemical burns of the eye are not time critical in terms of transport to hospital and irrigation should occur at the scene whenever feasible, because it is irrigation that makes a difference to the outcome. The method of irrigation must ensure cross contamination of an uninjured eye is avoided.
- Dressing burns is not a priority but may make burns less painful. Cling film should be applied after cooling and may provide some analgesia.
- Burn gels provide analgesia but are not a substitute for 20 minutes of cool running water. If burn gels are used these should be applied after cooling is complete.

Airway burns and inhalational injury

- Airway burns occur when there is direct thermal injury to the airway and this requires the patient to be enclosed in a small space (for example a car) or exposed for a significant period (for example trapped in a

house fire).

- It is common to see patients with superficial 'flash' burns to the face, for example from lighting a barbeque or lighting a fire with an accelerant. These burns often appear clinically concerning but they are not usually associated with airway burns, and patients rarely require intubation for airway protection in this setting.
- The signs of airway burns include:
  - Burn involving the inside of the mouth or nose.
  - Loss of nasal hair.
  - Stridor.
  - Voice change.
- Inhalational injury occurs when lung parenchyma is affected by chemicals within smoke. Inhalational injury does not usually require specific treatment in the out-of-hospital setting, other than general supportive treatment.
- The signs of inhalation injury include:
  - Hypoxia.
  - Respiratory distress.
  - Bronchospasm.
  - Black sputum.
- Bronchospasm following inhalational injury is not usually responsive to bronchodilators, however these may be administered using the **'exacerbation of asthma'** guideline, but do not administer magnesium or steroids.

Decontamination

- The most appropriate form of decontamination for a patient with chemical burns is a shower of approximately three minutes' duration, including a domestic shower. If heat has not been involved, the burn does not require cooling and in this setting the water should preferably be warm.
- Decontamination using a high pressure shower must be avoided if the patient has burns, as this may worsen the burn injury.
- Decontamination using a fire service hose (even on low pressure) should be avoided unless it is the only option, as this is likely to result in hypothermia.
- Decontamination using a fire service low pressure decontamination unit is acceptable, provided one is immediately available. However, decontamination should not be delayed waiting for one to arrive.

- The method of decontamination must ensure cross (or self) contamination is avoided.

Estimating burn depth and size

- Estimate burn depth only after cooling is complete:
  - Superficial burns do not have blisters and are red and painful like sunburn.
  - Partial thickness burns have blisters, weep fluid and are painful.
  - Full thickness burns are charred, white, leathery and usually painless.
- Estimate burn size only after cooling is complete:
  - Do not include superficial burns in the estimate of burn size.
  - The preferred method of estimating burn size is to use a piece of paper the same size as the patient's hand (including their fingers). This represents 1% of TBSA.
  - It is very easy to overestimate burn size.

Electrical injury

- Patients with burns following electrical injury may have significant muscle and nerve damage along the pathway of the current, even if the external burn area is small.
- Continuous ECG monitoring is required because dysrhythmia may occur.

Fluid loss from burns

- Shock as a result of fluid loss following burns takes several hours to develop.
- If a patient with burns has shock, look for an alternative cause.

Response and/or transport by helicopter

- Transport by helicopter may be indicated if the patient has burns that meet major trauma pathway to a regional burn facility (>5% in paediatric patients, >10% in adults), and it will save more than 60 minutes compared with road transport.
  - For helicopter transport criteria to be met, the road transport time will usually need to be more than 120 minutes.
- Transport or response by helicopter may be indicated if the patient has TBSA >30% or airway burns, and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or

- It will save more than 30 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

More Information

Version history

28 October 2024 1.0.5

Update: Multiple changes.